

Reproduce-then-Extend — Voter Turnout & Term Limits (Political Science) (Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Kuhlmann, R. & Lewis, D. C. (2017). “Legislative Term Limits and Voter Turnout.” *State Politics & Policy Quarterly* 17(4):372–392. Data: UNC Dataverse doi: 10.15139/S3/POGNLC. The original analysis is an **OLS** regression of state lower-chamber turnout on institutional predictors (Table 2, Col. 1). This example illustrates **out-of-sample prediction with a regression**.

1 Background

Kuhlmann and Lewis ask whether **legislative term limits shape voter turnout** in state legislative elections. Term limits reshape the electoral environment – forcing open-seat contests and changing candidate experience and the information available to voters – with theoretically ambiguous effects on participation. They use **OLS** to estimate the association between the strength of a state’s term limits and its lower-chamber turnout, holding constant the major institutional and demographic determinants of turnout (registration rules, professionalism, party competition, election timing, income, diversity). In the reproduced Table 2 model the **term-limitedness coefficient is positive and statistically significant** – more term-limited legislatures see modestly higher turnout – though the largest effects are presidential-election timing and ease of registration.

2 Data and codebook

Unit of analysis: a U.S. state lower-chamber legislative election (a **state-year**); $n = 486$.

Variable	Definition
turnout	lower-chamber election turnout rate, % of the voting-age population (outcome)
tlness	term-limitedness: strength/bindingness of legislative term limits
lcr2	log legislator-to-citizen ratio (per 100,000 residents)
mmds	share of legislative seats elected from multimember districts
regdifficulty	voter-registration closing date (days before the election)
straight	1 if the state offers straight-party-ticket voting
iuse2	log number of ballot initiatives on the ballot
squire	Squire index of legislative professionalism
ranney	Ranney index of interparty electoral competition
income2	per-capita income (thousands of dollars)
diversity3	Herfindahl index of racial/ethnic diversity
preselec	1 in presidential-election years

Variable	Definition
redistrict	1 if districts were newly redrawn (post-redistricting)
year	election year

3 Descriptive analysis

Before modeling, we profile the data thoroughly: the sample, the outcome, the predictors one at a time, missingness, each predictor's bivariate link to turnout, and the collinearity structure. This EDA motivates every choice in the reproduction that follows.

```
d <- read.delim(paste0(
  "https://raw.githubusercontent.com/bdesmarais/intro",
  "-ml-statistical-horizons-4day/main/tutorials/day1_",
  "intro_and_evaluation/01_turnout/data/turnout_state",
  ".tab"))
mvars <- c("turnout", "tlness", "lcr2", "mmds", "regdifficulty", "straight", "iuse2",
  "squire", "ranney", "income2", "diversity3", "preselec", "redistrict", "year")
d <- as.data.frame(na.omit(d[, mvars]))
preds <- setdiff(mvars, "turnout")

library(ggplot2)
library(dplyr)
library(tidyr)
library(e1071)
theme_set(theme_minimal(base_size = 9))

binvars <- c("straight", "preselec", "redistrict") # 0/1 indicators
numpred <- setdiff(preds, binvars) # continuous predictors
```

3.1 Dataset and provenance

The **unit of analysis** is a U.S. state lower-chamber legislative election (a state-year). After restricting to the modeling variables and dropping incomplete rows we have **n = 486** state-years and **14** variables: one continuous outcome (**turnout**), 10 continuous predictors, and 3 binary indicators. The data are Kuhlmann & Lewis's replication file (UNC Dataverse doi:10.15139/S3/POGNLC), assembling state turnout with institutional and demographic covariates across recent election cycles.

Table 1: Variable dictionary: role, type, and coding.

variable	role	type	description
turnout	outcome	continuous	lower-chamber turnout, % of voting-age population
tlness	predictor	continuous	term-limitedness (strength/bindingness of limits)
lcr2	predictor	continuous	log legislator-to-citizen ratio (per 100k)
mmds	predictor	continuous	share of seats from multimember districts
regdifficulty	predictor	continuous	registration closing date (days before election)
straight	predictor	binary	1 = straight-party-ticket voting offered
iuse2	predictor	continuous	log number of ballot initiatives
squire	predictor	continuous	Squire index of legislative professionalism
ranney	predictor	continuous	Ranney index of interparty competition
income2	predictor	continuous	per-capita income (\$1000s)
diversity3	predictor	continuous	Herfindahl index of racial/ethnic diversity
preselec	predictor	binary	1 = presidential-election year

variable	role	type	description
redistrict	predictor	binary	1 = newly redrawn (post-redistricting)
year	predictor	continuous	election year

3.2 The outcome in depth

Table 2: Turnout (%): summary statistics.

n	mean	sd	min	Q1	median	Q3	max	IQR	skew	kurt
486	43.67	10.52	14.67	36.13	42.72	51.85	74.6	15.73	0.05	-0.42

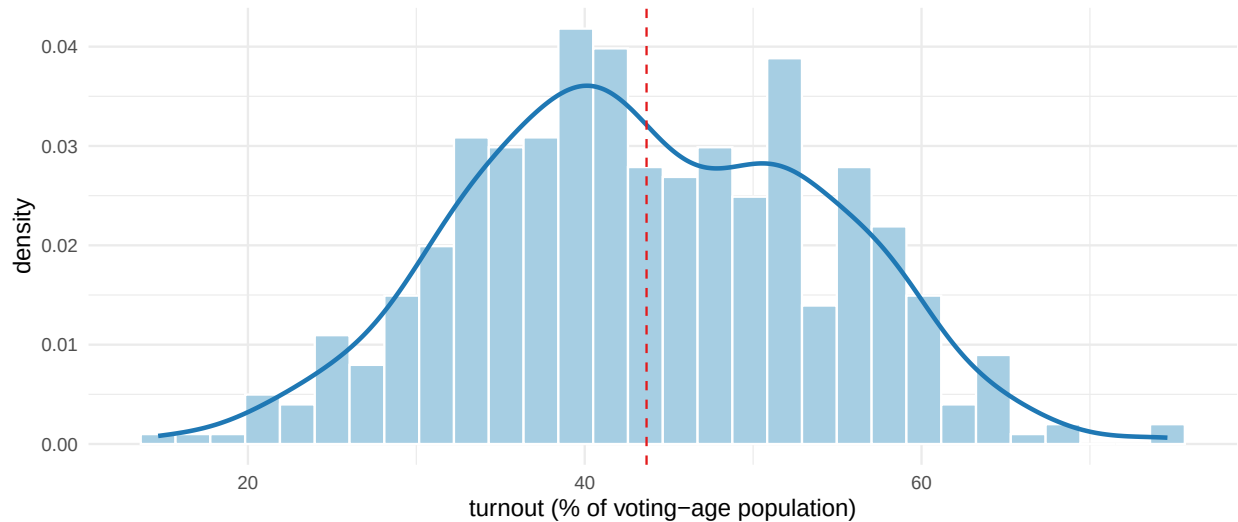


Figure 1: Distribution of state lower-chamber turnout, with density overlay and mean (dashed).

Turnout is close to **symmetric** (skewness 0.05) and mildly **platykurtic** (kurtosis -0.42, i.e. thinner tails than a normal), ranging from about 15% to 75% with no floor/ceiling pile-up and no extreme outliers. Because the distribution is already well-behaved, **no transformation of the outcome is warranted** — OLS on **turnout** in its natural units is appropriate.

3.3 Univariate predictor distributions

Table 3: Continuous predictors: summary statistics.

	n	n_miss	mean	sd	min	Q1	median	Q3	max	skew
tlness	486	0	0.08	0.28	0.00	0.00	0.00	0.00	1.57	4.13
lcr2	486	0	1.35	0.90	-1.05	0.77	1.28	1.94	3.61	0.16
mmds	486	0	0.16	0.34	0.00	0.00	0.00	0.00	1.00	1.91
regdifficulty	486	0	20.81	10.02	0.00	15.00	23.00	30.00	30.00	-0.96
iuse2	486	0	0.77	1.13	0.00	0.00	0.00	1.69	4.00	1.09
squire	486	0	0.20	0.13	0.05	0.12	0.16	0.23	0.66	1.67
ranney	486	0	0.89	0.09	0.63	0.83	0.91	0.96	1.00	-0.65
income2	486	0	27.89	8.74	12.53	20.82	26.85	33.98	56.96	0.51
diversity3	486	0	33.91	15.55	4.40	22.29	31.77	47.15	73.20	0.22

	n	n_miss	mean	sd	min	Q1	median	Q3	max	skew
year	486	0	1998.99	6.87	1988.00	1994.00	1998.50	2004.75	2010.00	0.01

Most predictors are moderately behaved; `squire` (professionalism) and `mmds` (multimember-district share) are right-skewed, reflecting a few highly professional legislatures and states leaning on multimember districts. The three binary indicators split as follows:

Table 4: Binary predictors: frequencies and share coded 1.

variable	n_0	n_1	pct_1
straight	303	183	37.7
preselec	262	224	46.1
redistrict	379	107	22.0

3.4 Missing-data analysis

Table 5: Missing values per variable in the modeling frame.

variable	n_missing	pct_missing
turnout	0	0
tlness	0	0
lcr2	0	0
mmds	0	0
regdifficulty	0	0
straight	0	0
iuse2	0	0
squire	0	0
ranney	0	0
income2	0	0
diversity3	0	0
preselec	0	0
redistrict	0	0
year	0	0

The modeling frame is built by `na.omit()` on the selected columns, so all 486 retained state-years are **complete cases** — every variable shows 0% missing above. Any incompleteness in the raw file was list-wise deleted up front; the reproduction therefore fits on a fully observed sample and no imputation is needed. (Because deletion happens before analysis, any bias would come from *which* state-years were dropped, not from missingness within the analysis set.)

3.5 Bivariate relationships with the outcome

For continuous predictors we correlate each with turnout; for the binary indicators we compare group means with a two-sample t-test.

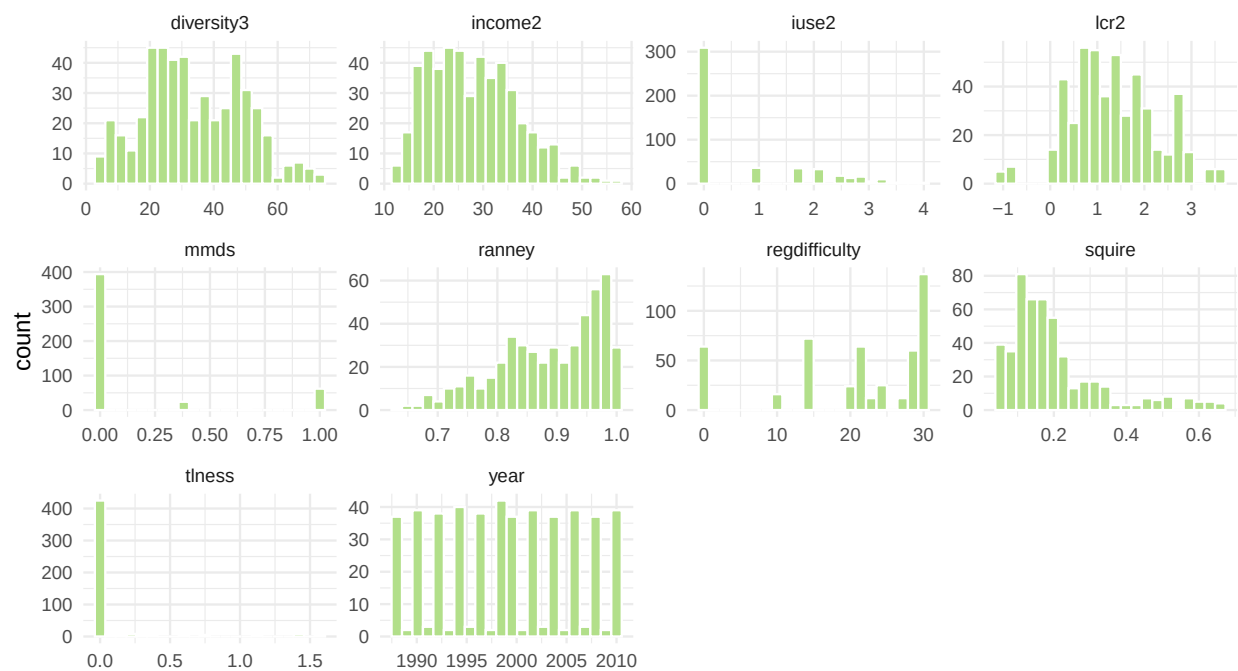


Figure 2: Distributions of the continuous predictors (faceted, free x-axes).

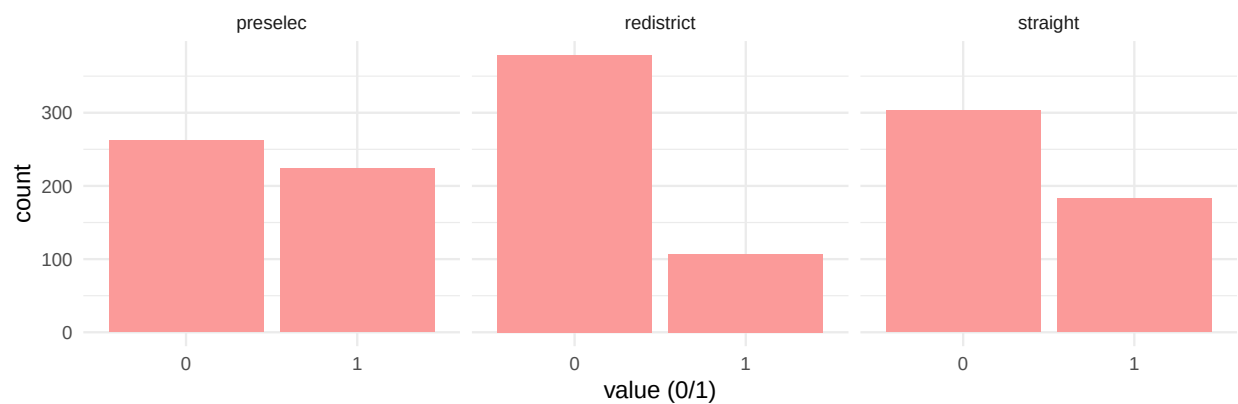


Figure 3: Class balance of the three binary predictors.

Table 6: Continuous predictors: Pearson r with turnout.

predictor	r_with_turnout
regdifficulty	-0.36
diversity3	-0.35
iuse2	0.29
lcr2	0.26
ranney	0.15
tlness	0.14
income2	0.08
mmds	-0.06
year	0.03
squire	0.03

Table 7: Binary predictors: turnout means by group (diff = 1 minus 0).

predictor	mean_1	mean_0	diff	p_value
straight	41.254	45.130	-3.876	0.000
preselec	50.885	37.502	13.383	0.000
redistrict	43.545	43.706	-0.160	0.891

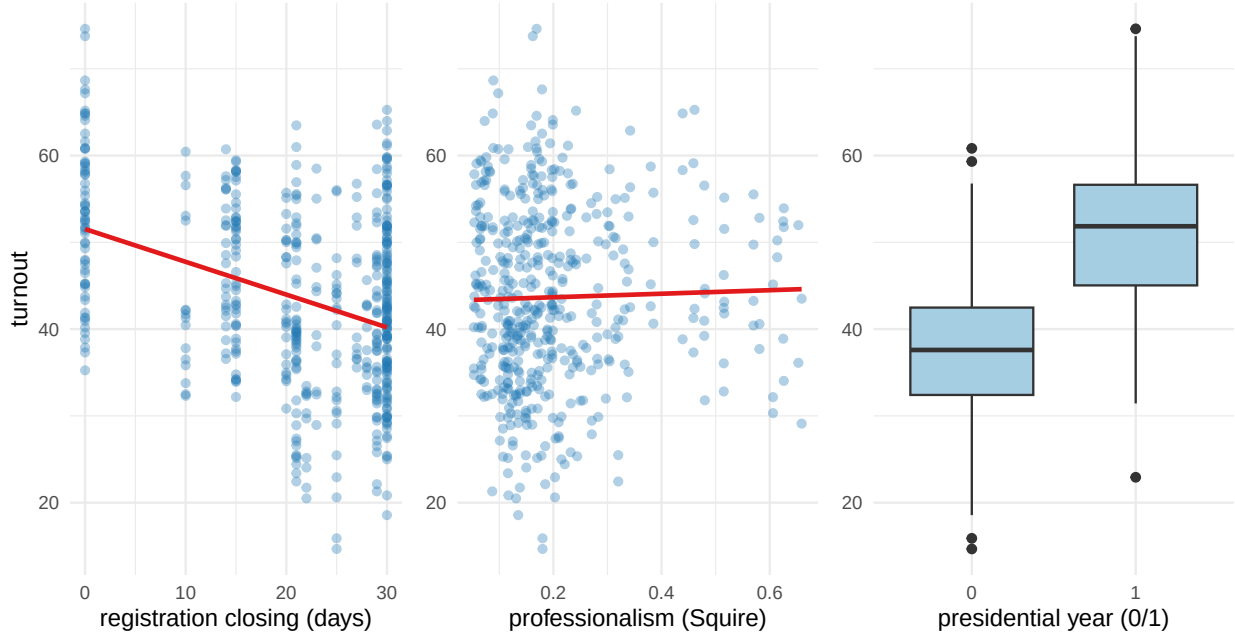


Figure 4: Turnout vs the two strongest continuous predictors (with linear fit), and by presidential-year timing.

The dominant marginal signal is **election timing**: presidential-year contests run far higher (see the **preselec** group difference), and easier registration (smaller **regdifficulty**) tracks higher turnout. Professionalism (**squire**) and competition (**ranney**) are positively associated, consistent with the paper’s story. **tlness** (term-limitedness) has only a modest marginal correlation — its role emerges once the institutional controls are held constant in the regression.

3.6 Multivariate structure and collinearity

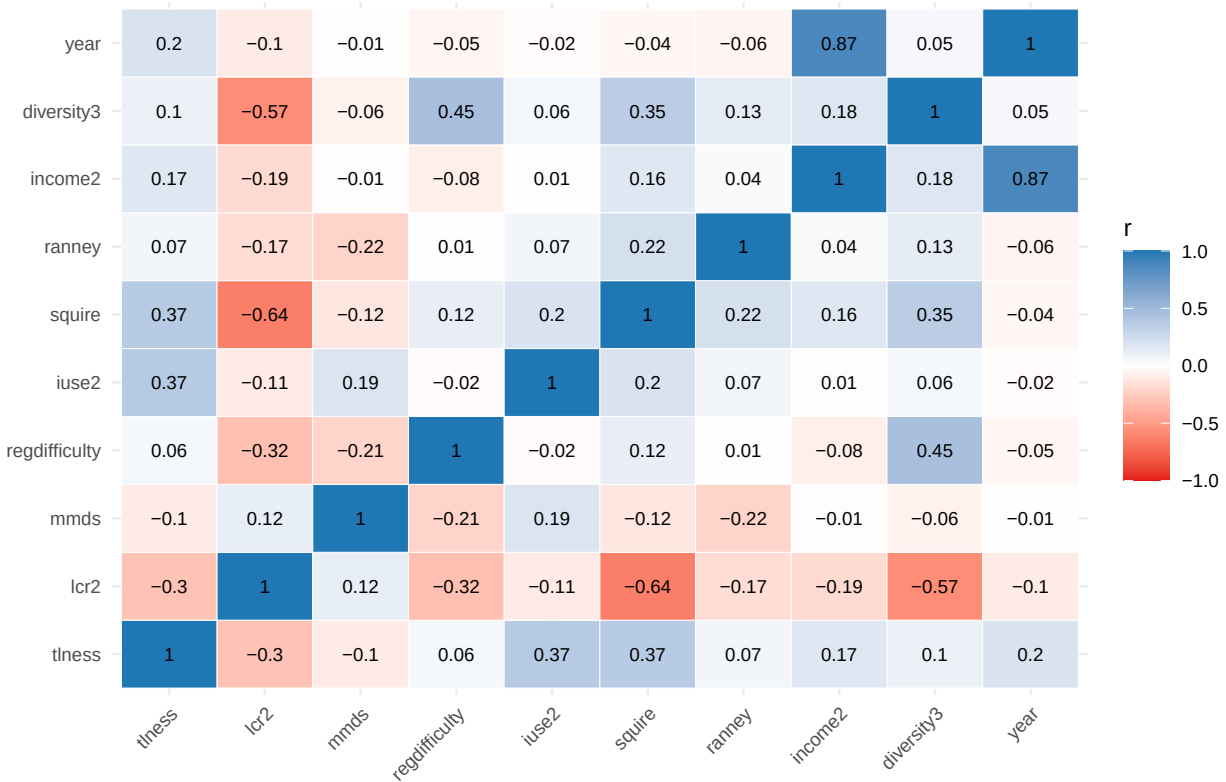


Figure 5: Correlation heatmap among continuous predictors.

Table 8: Variance inflation factors from the reproduction model.

predictor	VIF
income2	7.02
year	6.58
lcr2	2.48
squire	2.33
diversity3	1.89
straight	1.63
tlness	1.48
iuse2	1.46
regdifficulty	1.46
mmds	1.24
ranney	1.14
preselec	1.04
redistrict	1.04

One correlation exceeds the $|r| > 0.7$ flag: **income2** and **year** ($r = 0.87$) — per-capita income rises secularly over the sample window — and these two carry the largest VIFs (~ 7 and ~ 6.6). All other VIFs sit below 2.5. The collinearity is mild and confined to the income/time pair, so OLS coefficients on the substantive institutional predictors remain interpretable; it is worth noting only because it can inflate the standard errors on **income2** and **year** specifically. This motivates the regularized comparisons introduced in later tutorials, where correlated predictors are handled by shrinkage.

4 Reproduce the published regression

We fit their exact Table 2, Column 1 specification.

```
ols <- lm(turnout ~ tlness + lcr2 + mmds + regdifficulty + straight + iuse2 + squire +
           ranney + income2 + diversity3 + preselec + redistrict + year, data = d)
summary(ols)
```

```
##
## Call:
## lm(formula = turnout ~ tlness + lcr2 + mmds + regdifficulty +
##     straight + iuse2 + squire + ranney + income2 + diversity3 +
##     preselec + redistrict + year, data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20.1833  -3.7146   0.1051   4.0630  16.1266
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   78.69003   188.82397   0.417  0.67706
## tlness         2.24915    1.09331   2.057  0.04022 *
## lcr2           2.63156    0.44432   5.923 6.11e-09 ***
## mmds          -4.08496    0.82989  -4.922 1.18e-06 ***
## regdifficulty -0.24137    0.03068  -7.868 2.48e-14 ***
## straight     -1.01880    0.67165  -1.517  0.12997
## iuse2          2.20276    0.27242   8.086 5.24e-15 ***
## squire         9.69863    2.93923   3.300  0.00104 **
## ranney        13.29475    3.14016   4.234 2.76e-05 ***
## income2        0.22176    0.07727   2.870  0.00429 **
## diversity3    -0.13294    0.02256  -5.892 7.26e-09 ***
## preselec      12.78174    0.52126  24.521 < 2e-16 ***
## redistrict     1.44294    0.62707   2.301  0.02182 *
## year          -0.02805    0.09520  -0.295  0.76838
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.612 on 472 degrees of freedom
## Multiple R-squared:  0.723, Adjusted R-squared:  0.7154
## F-statistic: 94.79 on 13 and 472 DF, p-value: < 2.2e-16
```

Confirmation against the published study. Fitting their specification recovers the published model: 486 state-years and in-sample $R^2 = 0.723$. Presidential-election years raise turnout by ~12.8 points (**preselec**), stricter registration deadlines lower it (**regdifficulty**), and professionalism (**squire**) and competition (**ranney**) enter positively — reproducing the reported signs and significance.

5 Out-of-sample prediction with the regression

The same OLS that *explains* turnout can also *predict* it. We hold out a random 30% test set, fit on the other 70%, and evaluate on the unseen states — the core train/test workflow.

```
set.seed(2026)
train <- sample(nrow(d), floor(0.7 * nrow(d)))
test  <- setdiff(seq_len(nrow(d)), train)

fit   <- lm(turnout ~ ., data = d[train, ])
```



```

pred <- predict(fit, d[test, ])
actual <- d$turnout[test]

# out-of-sample error and R^2 (relative to the training-set mean)
rmse <- sqrt(mean((actual - pred)^2))
r2 <- 1 - sum((actual - pred)^2) / sum((actual - mean(d$turnout[train]))^2)
cat(sprintf("Train n=%d, test n=%d; out-of-sample RMSE=%.2f points, out-of-sample
  ↪ R^2=%.3f\n",
            length(train), length(test), rmse, r2))

## Train n=340, test n=146; out-of-sample RMSE=5.83 points, out-of-sample R^2=0.708

```

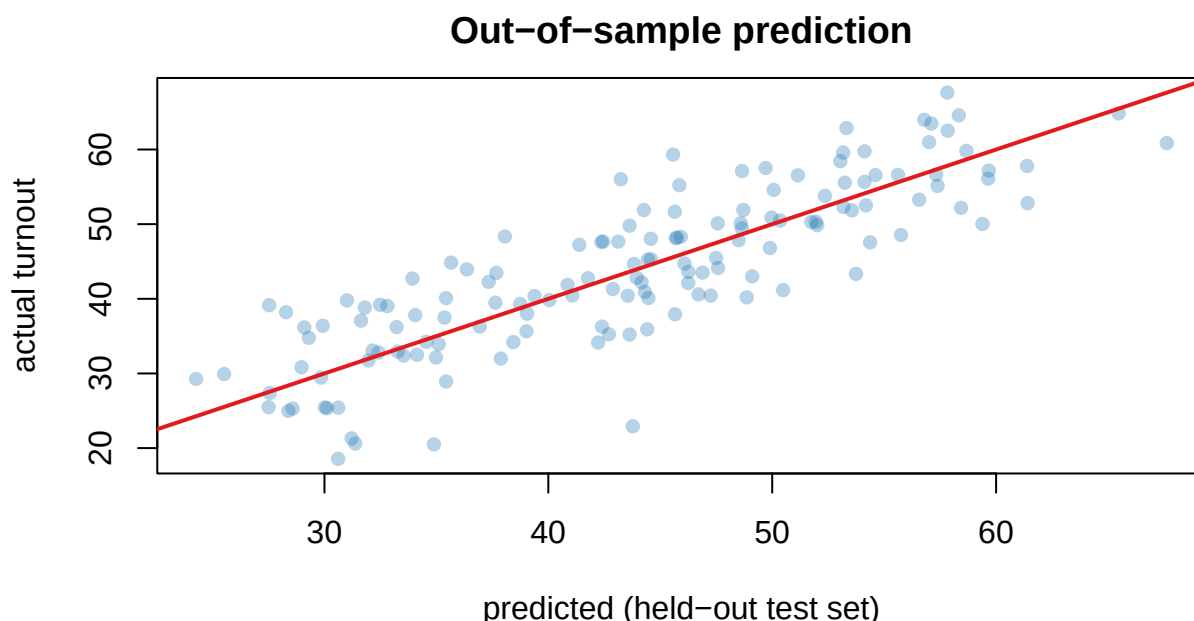


Figure 6: Predicted vs actual turnout on the held-out test set; the red line marks perfect prediction.

The held-out $R^2 \approx 0.71$ is the **honest** estimate of predictive accuracy — scored on data the model never saw, rather than the in-sample $R^2 = 0.72$ that always flatters the fit. The points cluster around the 45° line, so the regression predicts turnout in unseen states reasonably well.

6 Takeaway

The published OLS is the right tool for the paper’s *explanatory* question, and the *same model*, scored on a held-out test set, gives a calibrated estimate of how well turnout can be **predicted** out of sample. Explanation and prediction are different goals, and the train/test split is how we measure the second. (Later examples extend this baseline with polynomial terms, regularization, and tree-based learners.)

7 Recommended exercises

1. Re-run the train/test split under several random seeds (or switch to 5-fold cross-validation) and report the spread in out-of-sample RMSE. How stable is the estimate?
2. Add a quadratic term in `income2` (or an interaction between `preselec` and `regdifficulty`) and check whether held-out RMSE improves.
3. Fit a cross-validated lasso on the same predictors and compare its out-of-sample RMSE and its selected variables to OLS.